DAVID DUNN

Applied AI & Integration Engineer

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SUMMARY

Applied AI & Integration Engineer, 7+ years across data engineering, BI, and LLM-powered systems. Builds AI systems that plan, execute, review, and fix their own work, with a person approving what actually ships. Recent work: MCP servers and guarded computer use, budgeted retrieval with evaluation harnesses, multi-agent orchestration with human review gates, and voice-first RAG.

TECHNICAL PROJECTS

Heimdall · **Braid** · **Wayland Computer Use** — **Local-First Agent Runtime** — Go · Python · C · SQLite · MCP · Linux/Windows CI

- Built two MCP servers and an agent skill for guarded desktop control and observation (per-step focus/identity rechecks, failure-surviving execution ledger, accessibility-tree evidence); 27/27 paired E2E desktop tasks passed, batching cut median task time 37–47%, and change detection ran 2.58× faster with 99.8% smaller responses vs. screenshot polling; 167 tests.
- Built Braid, explainable budget-bounded retrieval fusing BM25, embeddings, graph PageRank, and recency, with an eval harness (grid search, no-answer cases, forbidden-result and budget gates); cut warm p50 query latency 19–41× at 2K–20K records.
- Built Heimdall, a Go task-continuity system for long-running human–agent work: event-sourced task state with replay, an MCP server with scoped credentials and explicit write grants, evidence-based completion checks, and reviewed state recovery; 293 tests.

Saga — **Multi-Agent Orchestration System** — Docker · MQTT · Gitea · Claude Code · Telegram · Proxmox

- Runs parallel Claude Code workers in sandboxed containers (no API keys, no host access) against Telegram spec intake, with critic review of each plan before code runs and self-healing PR loops.
- Human gates at every launch and merge: the system proposes, a person approves; agent roles, message-passing protocols, and review artifacts codified into reusable deployment patterns.

dbriefly — **Voice-First AI Audio Briefing Service** — Python · FastAPI · Postgres/pgvector · Next.js · Claude · OpenAI

- Built an end-to-end audio pipeline (Whisper ASR, pgvector embeddings, cross-source entity clustering, plan-and-execute LLM synthesis) turning 80+ podcasts, channels, newsletters, and forums into a daily 5–10 min brief with inline source clips at ~$0.03/brief.
- Shipped a realtime voice agent (OpenAI Realtime, ~500 ms budget) across web, Telegram, WhatsApp, SMS, and email on Postgres-only job orchestration, a per-call cost ledger, and a self-healing ops loop (automated critics + coding-agent fixes); ~700 integration tests.

Nexus — **Real-Time Prediction Market Intelligence** — Python · WebSockets · DuckDB · Claude · Convex · React

- Ingested 100K+ Kalshi/Polymarket events over 72h+ continuous runs (WebSocket streaming, integrity validation), surfaced through marketfinder, a live dashboard of 10K+ markets with anomaly alerts and trend clusters.
- Claude-powered semantic clustering, sliding-window anomaly detection, and cross-market correlation; pre-filtered 161M cross-platform pairs to ~1K before any LLM call (99.99% cost reduction); 158-test eval suite across 17 modules.

Also: **Skald** — Claude Code/Codex transcript archive (Go) · **steam-mcp** — open-source Steam Web API MCP server (TypeScript)

PROFESSIONAL EXPERIENCE

Applied AI & Integration Engineer | Dunn & Done Systems (independent) | Santa Clara, CA — **10/2024 – Present**

- Scoped and delivered Happily.love end-to-end for a matchmaking client: Claude-assisted onboarding interviews, invite-based applicant flow, structured profile model, approval workflows, and a private matchmaker dashboard for searchable profiles.
- Built reusable Python/Airflow ELT workflows for SMB clients, normalizing CRM, payment, ad, and accounting data into Redshift/dbt models with LLM-assisted metric validation and Metabase revenue/churn/LTV reporting; cut reporting effort 75%.

Analytics Engineer | Front Row | Remote — **10/2022 – 09/2024**

- Authored modular dbt models and Looker/Metabase explores answering 80% of recurring product-marketing questions, cutting ad-hoc requests 60%; owned metric definitions and quality rules.
- Built an experimentation pipeline (GTM, feature flags, dbt, Python uplift analysis) supporting 20+ A/B tests and dynamic-pricing tactics that lifted average purchase price 8%.

Marketing Data Analyst | ORComm | Seoul, South Korea — **01/2020 – 08/2022**

- Built reusable marketing dashboards and cohort/funnel/retention analyses for 10 seed-stage consumer startups, standardizing CAC, ROAS, and retention metrics across client reporting; reduced recurring report build time 60%.

Business Intelligence Analyst | VR Comet | Seoul, South Korea — **12/2018 – 12/2019**

- Transformed Google Analytics, HubSpot, and Unity event data in Redshift into reusable SQL views and reports for retention, monetization, and acquisition analysis, identifying growth opportunities tied to 15% MoM DAU growth.

TECHNICAL SKILLS

Applied AI & LLMs: Claude API, Claude Code (skills, plugins, hooks), OpenAI, Vertex AI, MCP servers (TS/Python/Go), hybrid RAG
Evals & Observability: eval harnesses, offline benchmarks, human-in-the-loop gates, cost ledgers, tracing, Grafana
Integration & Automation: FastAPI, Pydantic, Airflow, WebSockets, MQTT, event-driven systems, Docker, CI/CD, Playwright, Git, Linux
Languages & Data: Python, Go, TypeScript, SQL, dbt, Kafka, Spark, DuckDB, PostgreSQL, Snowflake, Redshift, BigQuery
Analytics, ML & BI: anomaly detection, A/B testing, forecasting, scikit-learn, statsmodels, Metabase, Looker, Tableau, Streamlit

EDUCATION

B.S. Physics, Lake Forest College (2009)